INCH-POUND

MIL-C-3098/130B
24 September 1992
SUPERSEDING
MIL-C-3098/130A
9 September 1977

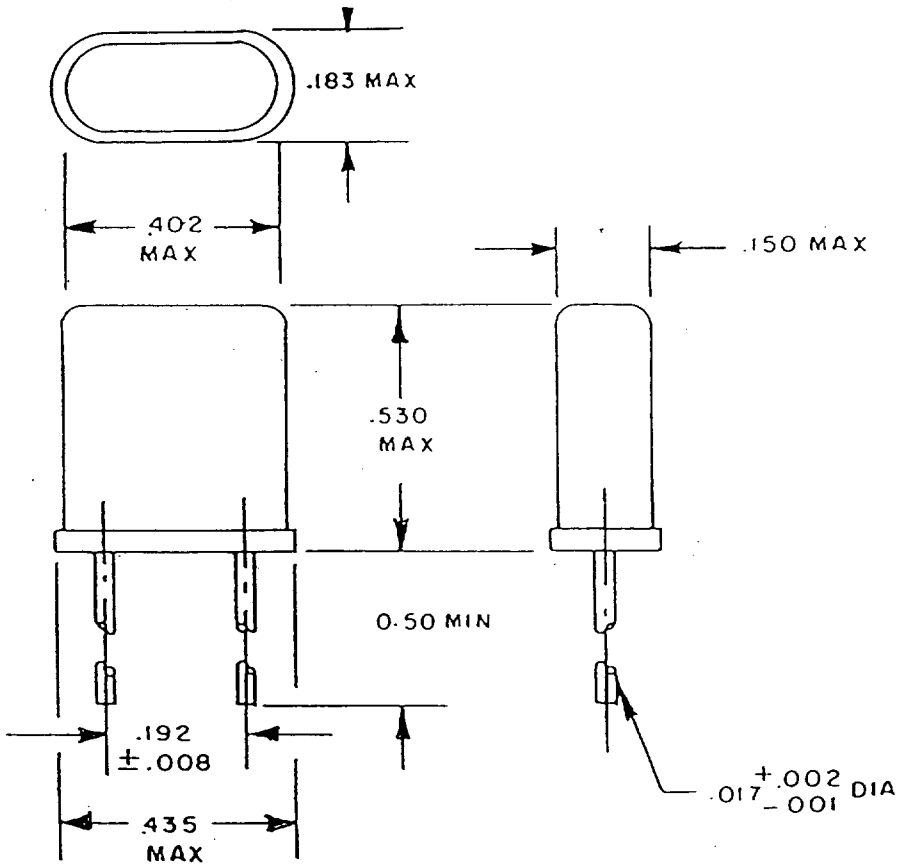
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR149/U

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 46 to 100 MHz; fifth mechanical overtone; noncontrolled; series resonance.



Inches	mm
.001	0.03
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR149/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/09).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 46.0 to 100 MHz, inclusive.

Operating temperature range:

Room temperature: 25°C to 30°C, inclusive.

Primary operating temperature range: 10°C to 25°C and 30°C to 80°C, inclusive.

Secondary operating temperature range : -40°C to +10°C and 80°C to 90°C, inclusive.

Equivalent resistance: 50 ohms.

Mode of oscillator: Fifth mechanical overtone.

Frequency tolerance:

Room temperature: ±10 ppm.

Primary: ±25 ppm.

Secondary: ±50 ppm.

Drive level: 2.0 mW maximum.

Capacitance, shunt: 6.0 +0.5, -1.0 pF.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: These crystal units are for use in the AN/ARC-101 equipment. They were previously fabricated using Bendix Corporation Specification 2088096-123 through 144.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0672-04)

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